

SIMATS ENGINEERING

ASSESSMENT TOOL 2

**COURSE NAME: OBJECT ORIENTED ANALYSIS AND
DESIGN COURSE CODE: CSA1102**

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 60%

QUESTIONS: 6 Total Marks:60 Annexure A

Application Based Questions

Code of Conduct:

*I Bheeshma L (192511332) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Bheeshma L

Annexure A

Application Based Questions

1. Use Case Diagram Analysis

A **Smart Hotel Reservation System** allows guests to search rooms, make reservations, cancel bookings, and process payments. Managers handle room allocation and reports, while payment services process transactions. The current design does not clearly distinguish actor responsibilities.

Analyze the system requirements, identify missing actors and use cases, and design an optimized Use Case Diagram showing proper relationships (include, extend, generalization) for better system clarity.

(10 Marks)

Answer:

a) Actors identified: Guest (primary actor who searches rooms, books, cancels, and pays), Hotel Manager (handles room allocation and generates occupancy reports), and Payment Gateway (external actor that processes transactions).

b) Primary use cases: Search Rooms, Make Reservation, Cancel Booking, Process Payment, Allocate Room, and Generate Occupancy Report.

c) Relationships: 'Make Reservation' and 'Cancel Booking' both include 'Login/Authenticate' since a guest must be verified first. 'Make Reservation' includes 'Process Payment', which itself includes 'Validate Payment Details' and extends to 'Send Confirmation' (an optional post condition step).

d) The Use Case Diagram below shows the Guest and Hotel Manager interacting with the system boundary, and the Payment Gateway interacting with the payment-related use cases, with include/extend relationships clearly marked.

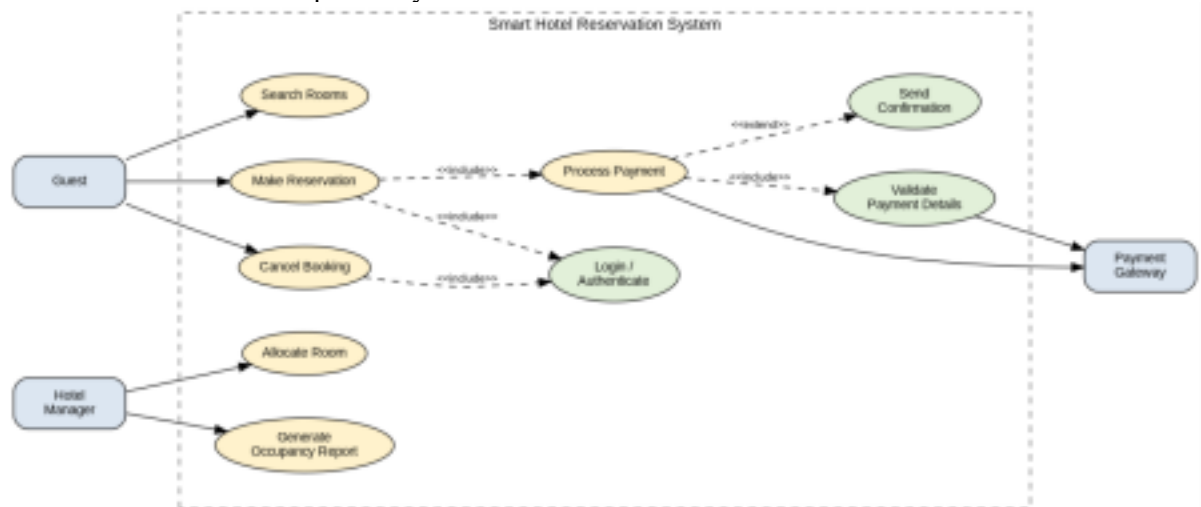


Fig 1: Use Case Diagram – Smart Hotel Reservation System

2. Class Diagram Design

A **University Management System** includes entities such as Student, Faculty, Course, Department, and Examination. The existing design lacks proper attributes, operations, and relationships among classes.

Analyze the system, identify relevant classes with attributes and methods, define relationships (association, aggregation, inheritance), and construct a well-structured Class Diagram.

(10 Marks)

Answer:

a) Classes with attributes and methods: Person (personId, name, email; getDetails()) is the base class. Student (rollNo, program, semester; enrollCourse(), viewResult()) and Faculty (facultyId, designation; assignCourse(), evaluate()) inherit from Person. Department (deptId, deptName; addFaculty(), addCourse()), Course (courseCode, title, credits; getSyllabus()), and Examination (examId, examDate, examType; scheduleExam(), publishResult()) complete the model.

b) Relationships: Inheritance – Student and Faculty generalize from Person. Aggregation – Department aggregates Faculty and Course (a department can exist without a specific faculty member/course). Association – Student is associated with Course (enrollment) and with Examination (appearing for exams); Course is associated with Examination (each course has exams).

c) Multiplicity: Department (1) – Faculty (1..*); Department (1) – Course (1..*); Student (*) – Course (*); Course (1) – Examination (1..*); Student (1) – Examination (*).

d) The Class Diagram below shows all classes with their attributes, operations, and the inheritance, aggregation, and association relationships with multiplicity.

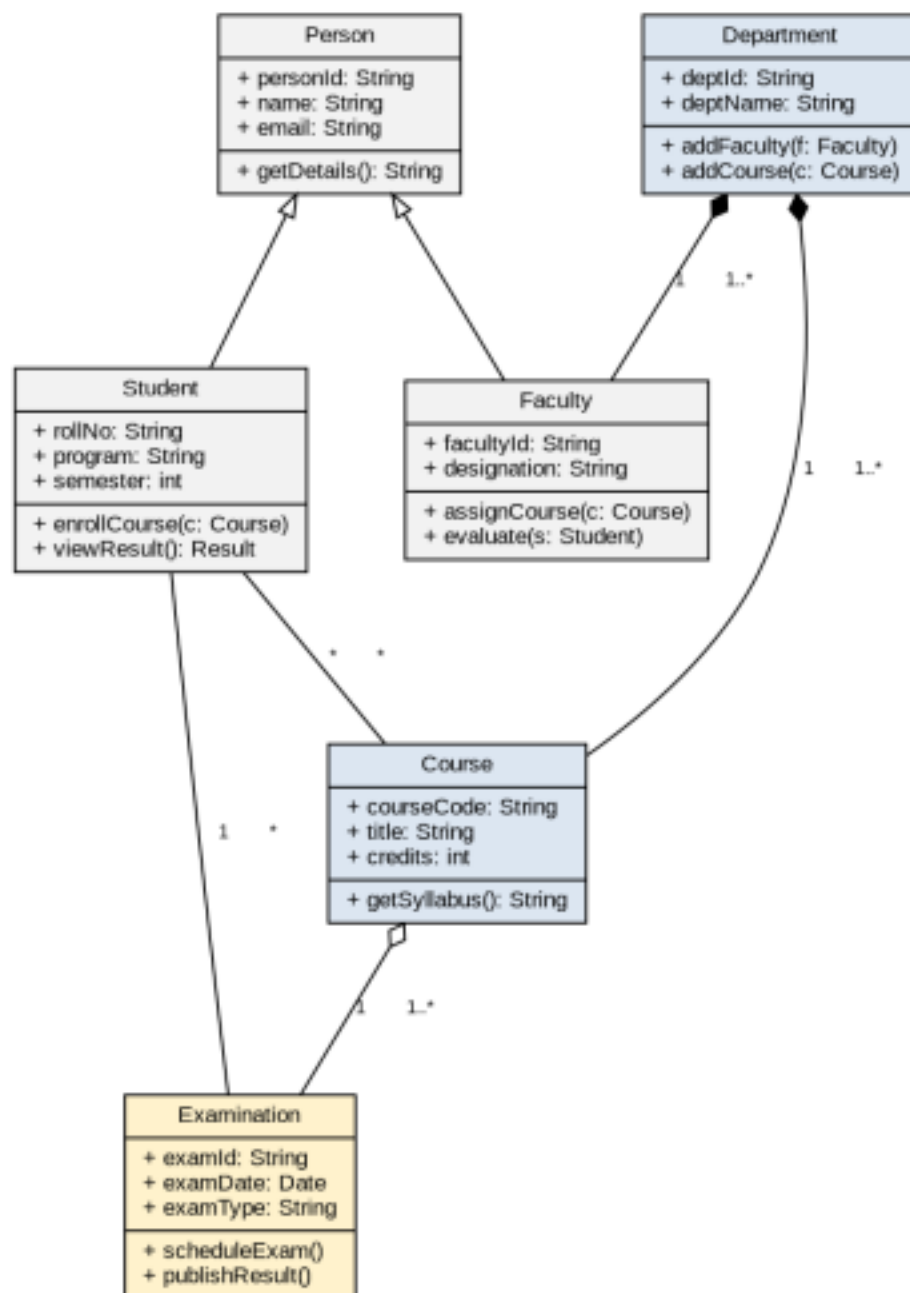


Fig 2: Class Diagram – University Management System

3. Sequence Diagram Evaluation

In a **Vehicle Rental System**, a customer searches for a vehicle, books it, makes payment, and receives a rental confirmation. The interaction among system components is incomplete and poorly documented.

Analyze the process flow, identify missing objects and message sequences, and design a detailed Sequence Diagram representing proper interaction among system components.

(10 Marks)

Answer:

- Objects involved: Customer, Search Interface, Rental System, Payment Gateway, and Confirmation Service.
- Message flow: The Customer sends `searchVehicle()` to the Search Interface, which requests `getAvailableVehicles()` from the Rental System and returns the list for display. The Customer then selects a vehicle and sends `bookVehicle()`, the Rental System checks availability internally, then calls `initiatePayment()` on the Payment Gateway, which validates the transaction and returns `paymentSuccess`. The Rental System finally calls `generateConfirmation()` on the Confirmation Service, which sends the `rentalConfirmation()` back to the Customer.
- Activation and lifelines: Each object has a dashed vertical lifeline; activation bars are shown on the Rental System (active throughout the booking process) and the Payment Gateway (active only during transaction validation).
- The Sequence Diagram below represents the complete interaction flow with numbered synchronous calls (solid arrows) and return messages (dashed arrows).

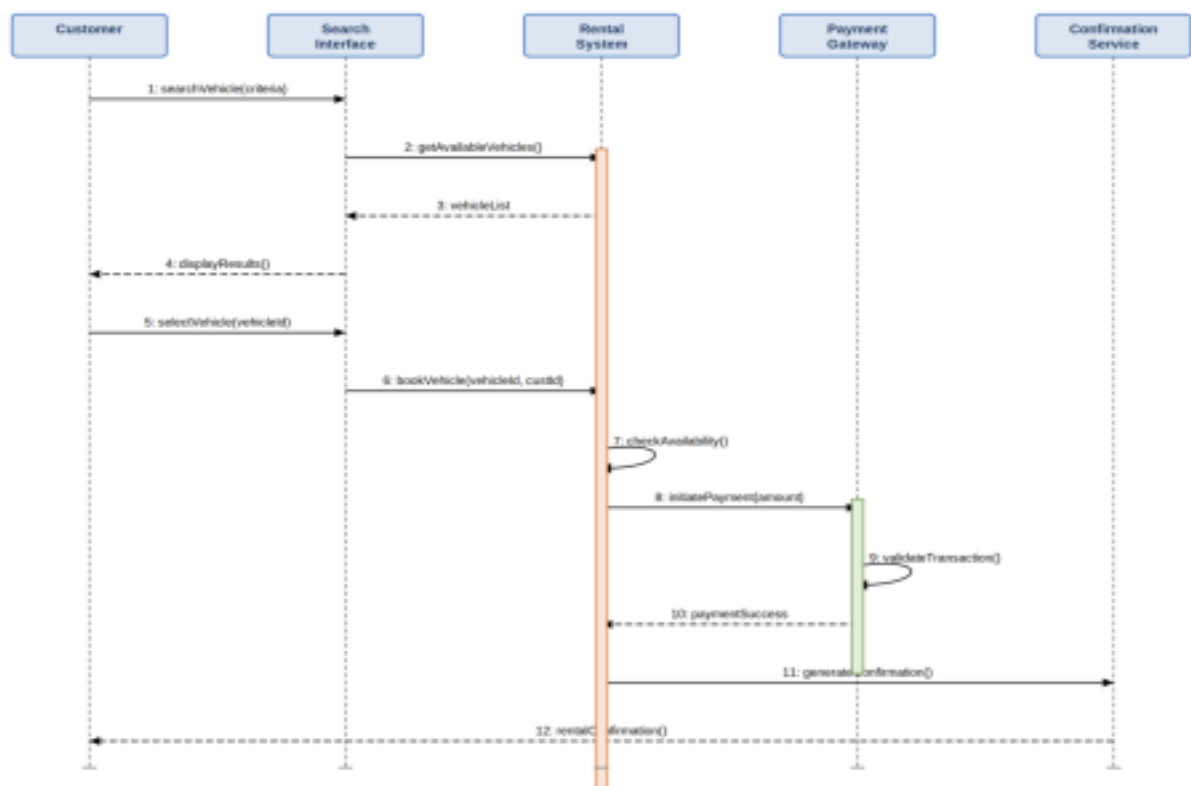


Fig 3: Sequence Diagram – Vehicle Rental System

4. State Diagram Construction

A **Traffic Signal Control System** operates through states such as Red Signal, Green Signal, Yellow Signal, Emergency Mode, and Maintenance Mode. The transitions between states are not clearly defined.

Analyze the system behavior, identify valid states and transitions, and construct a State Transition Diagram ensuring all events and conditions are properly represented.

(10 Marks)

Answer:

- States identified: Red Signal, Green Signal, Yellow Signal, Emergency Mode, and Maintenance Mode, along with an initial state (system start) and a final state (system shutdown).
- Events triggering transitions: timer expiry moves the signal through Red → Green → Yellow → Red in sequence; detection of an emergency vehicle transitions any active signal state into Emergency Mode, which returns to Green once the emergency is cleared; a maintenance trigger moves Red or Green into Maintenance Mode, which returns to Red once maintenance is completed or proceeds to the final state on system shutdown.
- Initial and final states: the black filled circle marks system start (entering Red Signal) and the double circle marks the final state reached after a system shutdown from Maintenance Mode.
- The State Transition Diagram below represents this behavior, with solid arrows for normal/emergency transitions and dashed arrows for maintenance-related transitions.



Fig 4: State Transition Diagram – Traffic Signal Control System

5. Component Diagram Design

A **Smart Healthcare Portal** consists of modules such as Patient Management, Appointment Scheduling, Medical Records, Billing, and Notification Services. The current architecture does not clearly define component interactions and dependencies.

Analyze the system architecture, identify major components and their interactions, and design a Component Diagram showing dependencies and interfaces between modules.

(10 Marks)

Answer:

a) Major components: API Gateway/Auth Interface, Patient Management, Appointment Scheduling, Medical Records, Billing, and Notification Services.

b) Interfaces: the API Gateway exposes an authenticated entry interface used by Patient Management; each downstream component exposes a service interface consumed by the components that depend on it (for example, Notification Services exposes an interface that Appointment Scheduling, Medical Records, and Billing all invoke).

c) Dependencies: Patient Management depends on Appointment Scheduling and Medical Records; Appointment Scheduling depends on Billing and Notification Services; Medical Records and Billing both depend on Notification Services.

d) The Component Diagram below shows these components with dashed dependency arrows (<<uses>>) representing the interface-based interactions.

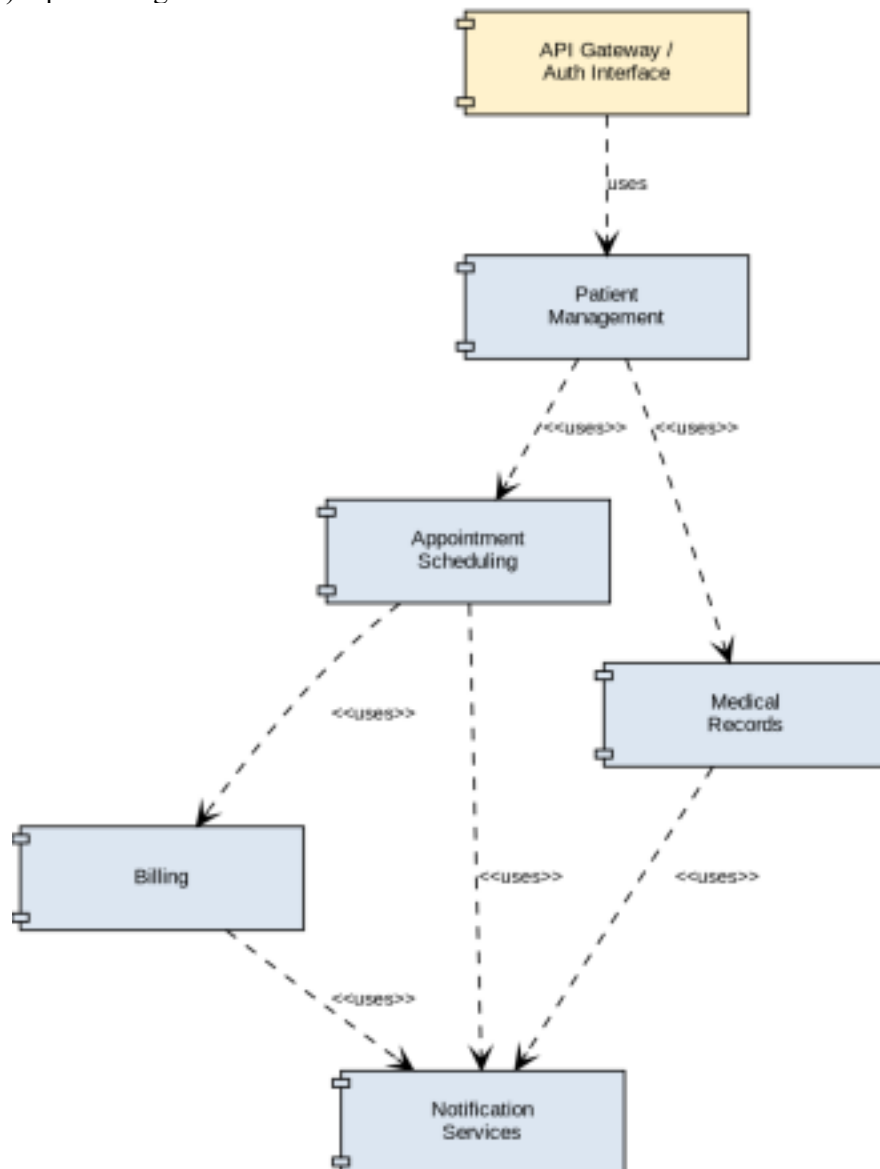


Fig 5: Component Diagram – Smart Healthcare Portal

6. Deployment Diagram Design

A **Smart City Waste Management System** is deployed across IoT waste-bin sensors, edge devices, cloud servers, and monitoring dashboards. The current design lacks clarity regarding physical architecture and communication links.

Analyze the deployment environment, identify nodes, artifacts, and communication links, and construct a detailed Deployment Diagram representing the complete system architecture.

(10 Marks)

Answer:

- Nodes identified: Waste-Bin Sensor Node (IoT device), Edge Gateway Device, Cloud Server, Monitoring Dashboard, and Collection Crew Device.
- Software artifacts deployed: the Sensor Node runs fill-level sensor firmware; the Edge Gateway runs a data-aggregator service; the Cloud Server hosts the Waste Management Service and its database; the Monitoring Dashboard runs the web dashboard application; the Collection Crew Device runs a mobile app.
- Communication links: sensors connect to the edge gateway using MQTT/LoRaWAN; the edge gateway forwards aggregated data to the cloud server over HTTPS/REST API; the cloud server pushes data to the monitoring dashboard over HTTPS and sends alerts to the collection crew's mobile app via HTTPS/push notifications.
- The Deployment Diagram below represents the complete physical architecture with nodes as 3-D boxes and communication links labelled with their protocols.

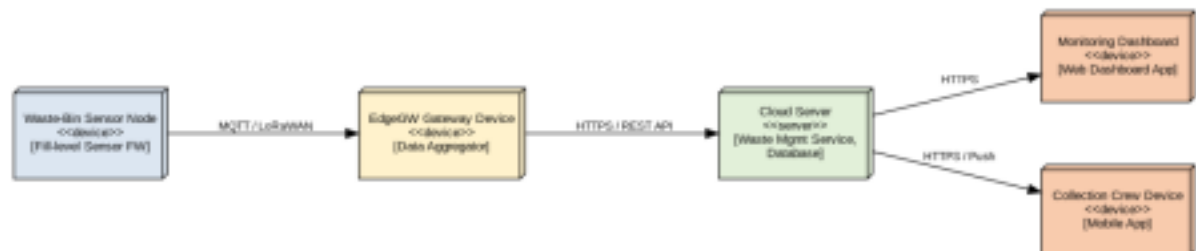


Fig 6: Deployment Diagram – Smart City Waste Management System

RUBRICS FOR CALCULATION

Application Based Questions

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem Solving Approach	20	Logical, well structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand